

Instructor: Date:

Learner: Regional Branch:

Program: Center:

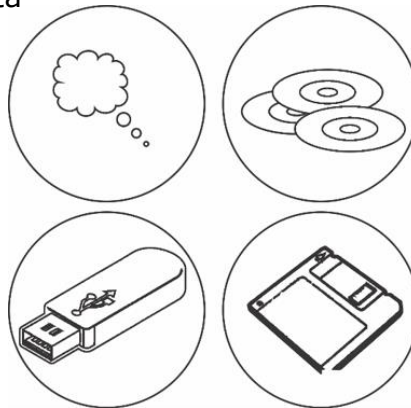
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Let's Start!

Different Ways to Store Data



Learning Outcome:

Build the user interface, supported on the prototype assessment, determining the inputs and outputs required in the design and defining the navigation guidelines, according to the user's needs.

Identifies different hardware components and peripheral devices.

Can Do Statements:

- I can talk about different types of storage.
- I can give information about optical storage.
- I can describe magnetic storage.

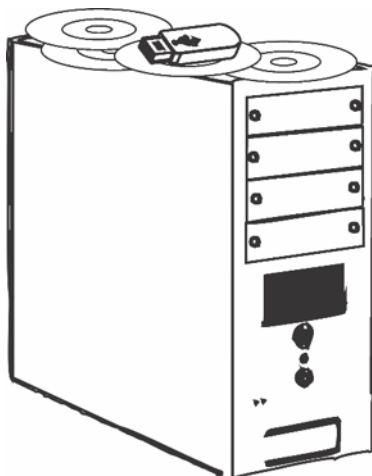
Ask your partner:

- Why do you think it is important to have a physical storage system in a computer?
- What type of storage systems do you trust the most to keep your files? Why?



Let's Explore!

Preparing a Presentation about Types of Storage!



Instructions: Read the text below.

Jason: I have to prepare a presentation about one type of storage, and I was wondering if you could help me. You are known for being an expert with computers.

Barb: Sure, what type of storage do you want to talk about?

Jason: I'm not sure. Can you talk to me a little about each type?

Barb: Well, I can think of four types. Magnetic storage, which is used to record and read digital content in a magnetic form; optical storage, which is used with discs and laser technologies; solid-state storage, which is about portable devices; and finally, cloud storage, which is related to an online service with a network memory that you can expand. All of them were designed as non-volatile memories due to the fact that volatile memories are intended to clear all data when the power gets off.

Jason: Can you give me more information about optical storage? That one sounds interesting.

Barb: First you need to understand the difference between disc and disk. Discs were created to read and write information through a laser. They can handle a high-capacity storage in different formats, and the data is more durable since it is difficult for them to record errors. Besides, the information is more secure. That's what I can tell you about optical storage.

Jason: And what is a disk?

Barb: Disks were made for magnetic storage. Magnetic storage has a large-capacity to file data provide rapid access to the user. Before, the most common device was the floppy disk but now we have hard drives, you can use them in laptops or desktop computers. Unfortunately magnetic storage devices can be easily destroyed or infected by virus.

Jason: I guess I will choose the optical storage then.





Let's Practice!

Barb knows everything about storage systems!

Instructions: Write the words to correctly complete the text. Use words from the box below.

perceived / created / used / write / recorded

Discs are _____ to read and _____ information through lasers. Similarly, different kinds of data are _____ on discs because they can last longer. However, due to the fact that discs _____ as magnetic drives, they are commonly _____ for storing all the data generated by the computer.



Let's Practice!

This is the description of a magnetic storage device!

Instructions: Select the correct answer.

1. Solid hard drives are magnetic storage devices due to the material they _____.
 a. were made of
 b. are impressed
 c. were making
 d. are driven
2. Initially, magnetic discs _____ for internal storage, but nowadays, they are also used as removable discs of high storage capacity.
 a. are implemented
 b. were implemented
 c. is implemented
 d. was implemented
3. Magnetic storage media was _____ with the only purpose of filing huge amounts of data, while providing rapid access to the user.
 a. assembled
 b. used by
 c. created
 d. implemented in
4. Since the beginning, external drives, solid drives, and hard drives _____ as magnetic storage drives due to their physical characteristics.
 a. were cataloged
 b. are bought
 c. are catalyzed
 d. were brought
5. Magnetic storage devices _____ to last longer than any other means of storage.
 a. is designed
 b. were designed
 c. was designed
 d. are designing





Extension Activity

Let's write about devices!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about optical and magnetic storage systems.
- Choose 3 devices of each one of them.
- Draw the devices that you have chosen.
- Write a description for each device and share your descriptions and pictures with your partners.

Self-Assessment

Yes**No**

- I search information about optical and magnetic storage. ☐ ☐
- I write a description for each device. ☐ ☐
- I share my product with my partners. ☐ ☐



Target Words

- **expand** (verb): to make something bigger in capacity
e.g. Cloud storage, which is related to an online service with a network memory that you can expand.
- **volatile** (adjective): a device that does not retain data when it is powered off
e.g. Volatile memories are intended to clear all data when the power gets off.
- **non-volatile** (adjective): a device that retains data even if it is powered off
e.g. All of them were designed as non-volatile memories.
- **floppy disk** (noun): plastic and portable device used to transport files
e.g. The most common device was the floppy disk but now we have hard drives.
- **format** (noun): the coding in which something is designed
e.g. They can handle a high-capacity storage in different formats.
- **desktop** (adjective): something that can fit on a desk
e.g. You can use them in laptops or desktop computers.





Learning Points

Simple Past Passive- (Affirmative Form.)

You use **Simple Past Passive** to provide more importance to the object and a past action in a sentence rather than the subject

Simple past Passive is used when the subject of the sentence can be omitted since it is implicit in the message. On the other hand, to mention the subject you have to use the preposition **by** followed by the noun.

Let's compare these two sentences:

Active voice:

- **People** used floppy disks to back up and transport data.

Passive voice:

- **Floppy disks** were used (by people) to back up and transport data.

As you can see, in passive voice the subject of the sentence "people" is not necessary; therefore, omitted.

Grammar structure for passive voice:

Object of the sentence + **Past of be** + Verb in past participle + Complement + (agent).

Example:

- Different types of storage devices **were** designed for individual needs.

Examples

- All of them were designed as non-volatile memories due to the fact that volatile memories are intended to clear all data when the power gets off.
- Discs were created to read and write information through a laser.
- Disks were made for magnetic storage.





Learning Points

Types of Storage.

There are different **types of storage** that can be classified into four categories. Magnetic, optical, solid state, and cloud storage.

In the past, floppy disks were very common as a mean of storage. However, their storage capacity was very limited and their structure was fragile. On the other hand, hard disks have been evolving with time, nowadays we can find portable units with a large amount of space.

CDs have also been reformed in order to obtain various benefits. We can find different formats such as CD-DA (Digital Audio), CD-ROM (for computer programs), CD-R (Recordable), and CD-RW (rewritable). Besides the previous mentioned, we also have devices such as flash drives or cloud drives on the internet that have allowed us to transport and access information easier.